



Survey and Biology of Mistletoe (*Viscum coloratum* Kom.) in Northern State, Sudan

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Abstract:

A survey was conducted in six selected areas in Marawi Locality during the winter season of 2022/2023 to determine host range, density and to study the biology of mistletoe parasite (*Viscum coloratum*). The six selected areas were, Elhamadab, Kasinger, Marawi, Elzooma, Tangasi and Elgurare. The results of this survey revealed that, mistletoe parasitism was mainly on guava, ardib, nabg, lemon, grapefruit and mango. The highest number of haustoria vitality of this parasite (5.00) occurred in nabg, while the lowest was recorded in mango (00.00). The highest number of branches vitality of this parasite (7.00) occurred in nabg, while the lowest was recorded in mango (0.00). The highest number of flowers vitality of this parasite (70.00) occurred in nabg, while the lowest was recorded in mango (0.00). The highest number of fruits and seeds vitality of this parasite (60.00) occurred in nabg, while the lowest was recorded in mango (0.00). Among the six hosts, the highest number of haustoria of this parasite (4.38) occurred in Elhamadab and Kasinger in acacia, while the lowest (3.00) was recorded in Elgurare, Elhamadab and Tangasi in guava. The highest number of branches of this parasite (7.38) occurred in Marawi in lemon, while the lowest (3.12) was recorded in Elhamadab in grapefruit. The highest number of flowers of this parasite (31.59) occurred in Elhamadab in acacia, while the lowest (3.21) was recorded in Tangasi in grapefruit. The highest number of fruits and seeds of this parasite (23.79) occurred in Elhamadab in acacia, while the lowest (1.68) was recorded in Elgurare in guava. In all six hosts, the mistletoe parasite gave two generations in the winter season and only one generation in the summer season (three generations/year). The number of generations which obtained in the winter season is more than which obtained in the summer season.

مستخلص :

أُجري المسح في ست مناطق مختارة في محلية مروحي خلال الموسم الشتوي للعام 2022/2023 لتحديد المدي العائلي، الكثافة، ولدراسة حيوية طفيل الهدال. المناطق الست المختارة كانت الحماداب، كاسنجر، مروحي، الزومة، تنقاسي والقرير. أظهرت نتائج هذا المسح أن تطفل طفيل الهدال كان علي الجوافة، العرديب، النبق، الليمون، القريب فروت والمانجو. أعلى رقم لحيوية الممصات لهذا الطفيل (5.00) ظهر في النبق، بينما الأقل سجل في المانجو (0.00). أعلى رقم لحيوية الفروع لهذا الطفيل (7.00) ظهر في النبق، بينما الأقل سجل في المانجو (0.00). أعلى رقم لحيوية الأزهار لهذا الطفيل (70.00) ظهر في النبق،

بينما الأقل سجل في المانجو. (0.00) أعلى رقم لحيوية الثمار والبذور لهذا الطفيل (60.00) ظهر في النبق، بينما الأقل سجل في المانجو. (0.00)

بين العوائل الستة، أعلى رقم للمصحات لهذا الطفيل (4.38) ظهر في الحماداب والكاسنجر في الطلح، بينما الأقل (3.00) سجل في القرير، الحماداب وتتقاسي في الجوافة. أعلى رقم للفروع لهذا الطفيل (7.38) ظهر في مروي في الليمون، بينما الأقل (3.12) سجل في الحماداب في القريب فروت.

أعلى رقم للأزهار لهذا الطفيل (31.59) ظهر في الحماداب في الطلح، بينما الأقل (3.21) سجل في تتقاسي في القريب فروت. أعلى رقم للثمار والبذور لهذا الطفيل (23.79) ظهر في الحماداب في الطلح، بينما الأقل (1.68) سجل في القريري الجوافة.

في كل العوائل الستة، أعطي طفيل الهدال جيلين في موسم الشتاء وجيل واحد فقط في موسم الصيف (ثلاثة أجيال في السنة). عدد الأجيال المتحصل عليه في موسم الشتاء أكثر من المتحصل عليه في موسم الصيف.

Introduction:

Intrusive flowering plants are a distinct group and different from the rest of the plant groups in their ways of life that they have adapted and are unique, meaning that they live on green plants (non-parasitic). The phenomenon makes flowering parasites a separate group that has its own characteristics that distinguish it from the rest of the world's groups. All flowering parasites belong to the group of angiosperms, and only one parasitic plant differs from this affiliation, which is *Parasitaxus*, to which it belongs to gymnosperms (Babiker, 2007).

Parasitic weeds are the plants which attack other plants by making connections and deriving part or all of their food from the hosts. They attach themselves either to the roots or the shoots of the host plant and survive on food material available in them. The parasitic weeds are host specific; they cannot survive in the absence of their host plant (Babiker, 2007). Mistletoe (*Viscum coloratum*) belongs to the Loranthaceae Family. It is flowering parasitic weed, partial parasitism and has been recorded on a huge number of deciduous hosts and is considered to have a wide host range, attacks many shoots of trees and shrubs by making connections and deriving part of its food from the hosts. Thus, infection by this parasite weed results in great losses. *V. coloratum* has higher transpiration rate than their hosts, an adaptation that ensures they always have a favorable water status relative to the host (Boussim *et al.*, 2004; Ejeta, 2007 and Jonathan and Gloucestershire, 2021). It has been used for a long time to beautify homes at Christmas and as known as the Christmas tree. This parasite parasitizes on the branches and stems of trees and shrubs (Boussim *et al.*, 2004). The absent work of past surveys and information regarding biology and survey of mistletoe in Northern State necessitates undertaking biology and survey of this parasitic weed to generate information on its density and distribution. The generated data help in understanding the size and extent of the problems that may arise due to this weed parasite and in developing management practices. The method used by Thomas (1985) is more effective in survey of weed species such as this parasite. A parasite survey was conducted in different six areas in Marawi Locality in Northern State to identify its distribution, abundance and incidence on various species of host, and to coordinate existing information on control measures and on conditions affecting their dispersal.

Materials and Methods:

The mistletoe (*V. coloratum*) survey was carried out during the winter season of 2022/2023 to identify its host range in various plants. This season period coincided with maximum growth of this parasite. Counts at this period may indicate the size and extent of parasite population. This survey was undertaken in six selected areas in Marawi Locality in Northern State that occupies the distant northern part of the Sudan and lies between latitudes 16°-22°N and longitudes 20°-32° E. The state lies in the arid and semi-arid (Suhair, 2017). The six selected areas were, Elhamadab, Kasinger, Marawi, Elzooma, Tangasi and Elgurare. Each of them more than 50 feddans (one fed. = 0.42 ha). Each of the study area was divided randomly into three sampling parts as A, B, and C based on infestation as suggested by Tizhe *et al.* (2016). The surveyed part in each area was divided into orchards or forests, of which 10 were randomly selected. In each selected orchards or forest, 10 trees or shrubs were randomly selected. The survey was undertaken using commonly accepted botanical survey methods. The survey methods involved searching and identifying this weed. The number of the mistletoe (*V. coloratum*) in different hosts was determined as suggested by Thomas (1985); Mohamed and Mohamed (1992) and Moeini *et al.* (2008). Thereafter, the environment was inspected so as to estimate the presence, abundance and level of infestation of the mistletoe in each location as suggested by Osman *et al.* (2007).

For mistletoe infested sites, distribution and severity data are used to assess management options as was used by Muir and Moody (2002). The biology experiment was conducted during two consecutive seasons (2021/ 2022 and 2022/2023) at Marawi Locality by putting seeds of this parasite on foliages of six hosts re-stitching guava, ardib, nabg, lemon, grapefruit, and mango to follow haustoria formation, their penetration to the different hosts, life cycle of this parasite and to determine number of its generations/year and number of its seeds/plant. The mean collected data was subjected to analysis of variance (ANOVA) and the means were separated using Duncan's Multiple Range Test (DMRT) as described by Gomez and Gomez (1984).

Results and Discussion:

A survey was conducted in six selected areas in Marawi Locality to determine host range, density and to study the biology of mistletoe parasite (*V. coloratum*). The six selected areas were, Elhamadab, Kasinger, Marawi, Elzooma, Tangasi and Elgurare. The six hosts were guava (*Psidium guajava*), Myrtaceae; ardib (*Tamarindus indica*) Fabaceae; nabg (*Ziziphus spina-christi*), Rhamnaceae; lemon (*Citrus Limon*), Rutaceae; grapefruit (*Citrus paradisi*), Rutaceae; and shittim wood (*Acacia seyal*), Mimosaceae. On guava host the highest number of haustoria of this parasite (3.90) occurred in Elhamadab, followed by Marawi (3.78), Kasinger (3.30), Elzooma (3.18), Tengasi (3.09), while the lowest was recorded in Elgurare (3.00). The mean number of haustoria in the six areas was (3.38). The highest number of branches of this parasite (6.00) occurred in Marawi, followed by Elhamadab (5.70), Kasinger (4.50), Elzooma (4.29), Tengasi (4.08), while the lowest was recorded in Elgurare (3.30). The mean number of branches in the selected six areas was (4.65). The highest number of flowers of this parasite (18.90) occurred in Elhamadab, followed by Kasinger (15.78), Marawi (14.79), Elzooma (14.19), and Tengasi (9.48), while the lowest was recorded in Elgurare (4.50). The mean number of flowers in the selected six areas was (12.93). The highest number of fruits and seeds of this parasite (10.29) occurred in Elhamadab, followed by Kasinger (8.88), Elzooma (6.48), Marawi (6.30), and Tengasi (3.96), while the lowest was recorded

in Elgurare (1.68). The mean number of fruits and seeds in the selected six areas was (6.27) as shown in Table (1).

Table 1: Biology of the mistletoe parasite in guava host in the six selected areas

Areas Treatments	Elha	Kas	Mar	Elz	Tan	Elgu	Mean
Number of haustoria	3.90	3.30	3.78	3.18	3.09	3.00	3.38c
Number of branches	5.70	4.50	6.00	4.29	4.08	3.30	4.65bc
Number of flowers	18.90	15.78	14.79	14.19	9.48	4.50	12.93a
Number of fruits	10.29	8.88	6.30	6.48	3.96	1.68	6.27b
Number of seeds	10.29	8.88	6.30	6.48	3.96	1.68	6.27b
C. V. %	-	-	-	-	-	-	30.21
SE±	-	-	-	-	-	-	0.83

Elha = Elhamadab; Kas= Kasinger; Mar = Marawi; Elz = Elzooama; Tan = Tangasi; and Elgu = Elgurare

Means with the same letters in the same column are not significantly different at 5% level of probability according to Duncan's Multiple Range Test (DMRT).

On lemon host the highest number of haustoria of this parasite (3.30) occurred in Kasinger, Marawi, and Elzooama, followed by Tengasi (3.18), Elgurare (3.09), while the lowest was recorded in Elhamadab (3.00). The mean number of haustoria in the selected six areas was (3.20). The highest number of branches of this parasite (7.38) occurred in Marawi, followed by Elhamadab (6.90), Kasinger (6.90), Elzooama (5.88), Tangasi (3.90), while the lowest was recorded in Elgurare (3.78). The mean number of branches in the selected six areas was (5.29). The highest number of flowers of this parasite (21.48) occurred in Elhamadab, followed by Kasinger (17.79), Marawi (14.19), Elzooama (12.18), and Tengasi (10.08), while the lowest was recorded in Elgurare (6.90). The mean number of flowers in the selected six areas was (13.77). The highest number of fruits and seeds of this parasite (16.29) occurred in Elhamadab, followed by Kasinger (14.88), Marawi (12.39), Elzooama (9.78), and Tengasi (8.49), while the lowest was recorded in Elgurare (6.18). The mean number of fruits and seeds in the selected six areas was (11.34). The mean numbers of flowers, fruits and seeds significantly increased as compared to the mean number of haustoria and branches as reported in Table (2).

Table 2: Biology of the mistletoe parasite in lemon host in the six selected areas

Areas Treatments	Elha	Kas	Mar	Elz	Tan	Elgu	Mean
Number of haustoria	3.00	3.30	3.30	3.30	3.18	3.09	3.20b
Number of branches	6.90	6.90	7.38	5.88	3.90	3.78	5.29bc
Number of flowers	21.48	17.79	14.19	12.18	10.08	6.90	13.77a
Number of fruits	6.29	14.88	12.39	9.78	8.49	6.18	11.34a
Number of seeds	16.29	14.88	12.39	9.78	8.49	6.18	11.34a
C. V. %	-	-	-	-	-	-	26.96
SE±	-	-	-	-	-	-	0.99

Elha = Elhamadab; Kas= Kasinger; Mar = Marawi; Elz = Elzooama; Tan = Tangasi; and Elgu = Elgurare

Means with the same letters in the same column are not significantly different at 5% level of probability according to Duncan's Multiple Range Test (DMRT).

On grapefruit host the highest number of haustoria of this parasite (3.30) occurred in Elhamadab, followed by Kasinger (3.18), Marawi (3.18), while the lowest was recorded in Elzoooma (3.09), Tengasi (3.09), and Elgurare (3.09). The mean number of haustoria in the selected six areas was (3.16). The highest number of branches of this parasite (3.90) occurred in Kasinger and Marawi, followed by Elzoooma (3.78), Tangasi and Elgurare (3.69), while the lowest was recorded in Elhamadab (3.12). The mean number of branches in the selected six areas was (3.68). The highest number of flowers of this parasite (12.69) occurred in Elhamadab, followed by Marawi (9.90), Kasinger (9.24), Elzoooma (7.68), Elgurare (4.59), while the lowest was recorded in Tangasi (3.21). The mean number of flowers in the selected six areas was (7.89). The highest number of fruits and seeds of this parasite (9.90) occurred in Marawi followed by Elhamadab (9.12), Kasinger (8.68), Elzoooma (6.48), and Tengasi (4.38), while the lowest was recorded in Elgurare (3.78). The mean number of fruits and seeds in the selected six areas was (7.06). The mean numbers of flowers, fruits and seeds significantly increased as compared to the mean numbers of haustoria and branches as shown in Table (3).

Table 3: Biology of the mistletoe parasite in grapefruit host in the six selected areas

Areas Treatments	Elha	Kas	Mar	Elz	Tan	Elgu	Mean
Number of haustoria	3.30	18 3.	3.18	3.09	3.09	3.09	3.16b
Number of branches	3.12	3.90	3.90	3.78	3.69	3.69	3.68b
Number of flowers	12.69	9.24	9.90	7.68	3.21	4.59	7.89a
Number of fruits	9.12	8.68	9.90	6.48	4.38	3.78	7.06a
Number of seeds	9.12	8.68	9.90	6.48	4.38	3.78	7.06a
C. V. %	-	-	-	-	-	-	29.41
SE±	-	-	-	-	-	-	0.69

Elha = Elhamadab; Kas= Kasinger; Mar = Marawi; Elz = Elzoooma; Tan = Tangasi; and Elgu = Elgurare

Means with the same letters in the same column are not significantly different at 5% level of probability according to Duncan's Multiple Range Test (DMRT).

On nabg host, the highest number of haustoria of this parasite (4.36) occurred in Elhamadab, followed by Kasinger (4.29), Marawi (4.24), Elzoooma (4.08), Tangasi (3.78) while the lowest was recorded in Elgurare (3.69). The mean number of haustoria in the selected six areas was (4.07). The highest number of branches of this parasite (6.90) occurred in Elhamadab and Marawi, followed by Elzoooma (6.69), Tangasi (6.48), Elgurare (6.18), while the lowest was recorded in Kasinger (6.15). The mean number of branches in the selected six areas was (6.55). The highest number of flowers of this parasite (24.18) occurred in Elhamadab followed by Kasinger (22.38), Marawi (15.30), Elzoooma (14.79), and Tengasi (9.27), while the lowest was recorded in Elgurare (8.49). The mean number of flowers in the selected six areas was (15.74). The highest number of fruits and seeds of this parasite (23.49) occurred in, Elhamadab followed by Kasinger (21.00), Marawi (15.18), Elzoooma (13.24), and Tengasi (9.18), while the lowest was recorded in Elgurare (4.68). The mean number of fruits and seeds in the selected six areas was (14.46). The mean numbers of flowers,

fruits and seeds significantly increased as compared to the mean numbers of haustoria and branches (Table 4).

Table 4: Biology of the mistletoe parasite in nabg host in the six selected areas:

Areas Treatments	Elha	Kas	Mar	Elz	Tan	Elgu	Mean
Number of haustoria	4.36	4.29	4.24	4.08	3.78	3.69	4.07b
Number of branches	6.90	6.15	6.90	6.69	6.48	6.18	6.55bc
Number of flowers	24.18	22.38	15.30	14.79	9.27	8.49	15.74a
Number of fruits	23.49	21.00	15.18	13.24	9.18	4.68	14.46a
Number of seeds	23.49	21.00	15.18	13.24	9.18	4.68	14.46a
C. V. %	-	-	-	-	-	-	33.41
SE±	-	-	-	-	-	-	1.51

Elha = Elhamadab; Kas= Kasinger; Mar = Marawi; Elz = Elzooama; Tan = Tangasi; and Elgu = Elgurare

Means with the same letters in the same column are not significantly different at 5% level of probability according to Duncan's Multiple Range Test (DMRT).

On orange host the highest number of haustoria of this parasite (3.30) occurred in Kasinger, followed by Elhamadab (3.18), Marawi (3.09), Elzooama (3.09), Elgurare (3.09), while the lowest was recorded in Tangasi (3.00). The mean number of haustoria in the selected six areas was (3.13). The highest number of branches of this parasite (3.78) occurred in Kasinger, Marawi, Elgurare followed by Elzooama (3.69), and Tangasi (3.60), while the lowest was recorded in Elhamadab (3.12). The mean number of branches in the selected six areas was (3.63). The highest number of flowers of this parasite (12.30) occurred in Elhamadab followed by Elzooama (6.90), Kasinger (6.48), Marawi (6.39), Tengasi (6.09), while the lowest was recorded in Elgurare (3.27). The mean number of flowers in the selected six areas was (6.91). The highest number of fruits and seeds of this parasite (9.90) occurred in, Elhamadab followed by Elzooama (4.68), Tengasi (3.24), Marawi (3.21), while the lowest was recorded in Elgurare (3.18). The mean number of fruits and seeds in the selected six areas was (4.77). The mean number of flowers significantly increased as compared to the mean numbers of other treatments as pointed in Table (5).

Table 5: Biology of the mistletoe parasite in orange host in the six selected areas

Areas Treatments	Elha	Kas	Mar	Elz	Tan	Elgu	Mean
Number of haustoria	3.18	3.30	3.09	3.09	3.00	3.09	3.13b
Number of branches	3.12	3.78	3.78	3.69	3.60	3.78	3.63b
Number of flowers	12.30	6.48	6.39	6.90	6.09	3.27	6.91a
Number of fruits	9.90	4.38	3.21	4.68	3.24	3.18	4.77b
Number of seeds	9.90	4.38	3.21	4.68	3.24	3.18	4.77b
C. V. %	-	-	-	-	-	-	34.20
SE±	-	-	-	-	-	-	1.28

Elha = Elhamadab; Kas= Kasinger; Mar = Marawi; Elz = Elzooama; Tan = Tangasi; and Elgu = Elgurare

Means with the same letters in the same column are not significantly different at 5% level of probability according to Duncan's Multiple Range Test (DMRT).

On acacia host the highest number of haustoria of this parasite (4.38) occurred in Elhamadab, Kasinger, followed by Marawi (3.90), Elzooma (3.69), Tangasi (3.60), while the lowest was recorded in Elgurare (3.30). The mean number of haustoria in the selected six areas was (3.88). The highest number of branches of this parasite (6.09) occurred in Elhamadab, followed by Kasinger (5.79), Marawi (5.28), Elzooma (5.19), Tangasi (4.89), while the lowest was recorded in Elgurare (4.68). The mean number of branches in the selected six areas was (5.32). The highest number of flowers of this parasite (31.59) occurred in Elhamadab followed by Kasinger (24.18), Marawi (20.58), Elzooma (15.78), Tengasi (12.90), while the lowest was recorded in Elgurare (12.69). The mean number of flowers in the selected six areas was (19.62). The highest number of fruits and seeds of this parasite (23.79) occurred in, Elhamadab followed by Kasinger (20.49), Marawi (19.59), Elzooma (14.58), Tengasi (12.00), while the lowest was recorded in Elgurare (11.19). The mean number of fruits and seeds in the selected six areas was (16.94). The mean numbers of flowers, fruits and seeds significantly increased as compared to the mean numbers of other treatments as mentioned in Table (6).

Table 6: Biology of the mistletoe parasite in acacia host in the six selected areas

Areas Treatments	Elha	Kas	Mar	Elz	Tan	Elgu	Mean
Number of haustoria	4.38	4.38	3.90	3.69	3.60	3.30	3.88b
Number of branches	6.09	5.79	5.28	5.19	4.89	4.68	5.32bc
Number of flowers	31.59	24.18	20.58	15.78	12.90	12.69	19.62a
Number of fruits	23.79	20.49	19.59	14.58	12.00	11.19	16.94a
Number of seeds	23.79	20.49	19.59	14.58	12.00	11.19	16.94a
C. V. %	-	-	-	-	-	-	25.01
SE±	-	-	-	-	-	-	1.28

Elha = Elhamadab; Kas= Kasinger; Mar = Marawi; Elz = Elzooma; Tan = Tangasi; and Elgu = Elgurare

Means with the same letters in the same column are not significantly different at 5% level of probability according to Duncan's Multiple Range Test (DMRT).

Among the six hosts, the highest number of haustoria of this parasite (4.38) occurred in Elhamadab and Kasinger in acacia host, while the lowest (3.00) was recorded in Elgurare, Elhamadab and Tangasi in guava host. The highest number of branches of this parasite (7.38) occurred in Marawi in lemon host, while the lowest (3.12) was recorded in Elhamadab in grapefruit host. The highest number of flowers of this parasite (31.59) occurred in Elhamadab in acacia host, while the lowest (3.21) was recorded in Tangasi in grapefruit host. The highest number of fruits and seeds of this parasite (23.79) occurred in Elhamadab in acacia host, while the lowest (1.68) was recorded in Elgurare in guava host. These previous mentioned results showed that, in this study acacia host has

the highest percentage of infestation with this parasite while guava and mango hosts are the lowest percentage of infestation. This could be attributed to the mentioned reasons by Aukema and Del Rio (2002) re-stitching seed dispersing birds favored larger infected trees as perching and feeding sites and deposited seeds disproportionately on them, which contributed to the mistletoe distribution being significantly aggregated within its host population. The research of Boussim et al. (2004) revealed that, host trees with large basal area have influence on mistletoes appearance and infestation on their host tree, Martinez et al. (1995) and Didier et al. (2009) opined that, mistletoes host preference was due to wide canopy spread and height of the host trees. These findings were contrary to the research of Aukema and Del Rio (2002) and Rexburg and Nikoloson (2008) who reported weak relationship between large host size and infection intensity of African mistletoes. Also these results could be due to feeding preference of the pollinator birds to acacia than host guava and mango, which visiting the area in search of nectar and nesting place. In the study area, trees of economic importance such as acacia were found to have high prevalence of occurrence by African mistletoes while guava and mango were having low prevalence of occurrence. Similar results were found by Anka, and Aliero (2019) who indicated that, acacia had the highest infestation of African mistletoe in Sokoto metropolis. The high infestation of this hemi parasite in Location Elhamadab area may be due to availability of water; this was supported by the findings of Ismailia (2000), who sited nearness to water source as an important factor influencing mistletoe distribution locally. Anka and Aliero (2019) also reported similar results in UDUS main campus, having closer proximity to both River Sokoto and River Rima, showed the highest percentage mistletoe compared to Gagi, which is located far away from rivers or similar water bodies had the least mistletoe infestation in Sokoto metropolis. The infestation on these trees could be due to wild distribution of seeds of mistletoes in the study area by broad array of birds that depend on these mistletoes for food. They consume the seeds of the mistletoes, thereby enhancing the transfer of the sticky seed from one tree to another (Van Ommeren and Whitham, 2002). Some of these trees with high infestation were observed to shed leaves completely during dry season, as they struggle to survive, this could lead to death and reduction of these trees in future. This is in congruence with the research of Begho *et al.* (2007); Ibrahim and Ayodele (2011) and Asare-Bediako *et al.* (2013) who indicated that, heavy infestation of African mistletoes in many parts of West Africa leading to gradual death and extinction of economic trees. Moreover, severe infested trees are prone to infection by other pathogenic agents and insect attack which contribute to the reduced growth and production of the trees in the study area. Despite the large host range in the study area, many mistletoes showed high preference to some host compared to the others.

In all six hosts, the mistletoe parasite gave two generations in the winter season and only one generation in the summer season (three generations/year). The number of generations which obtained in the winter season is more than which obtained in the summer season. This result could be attributed to the low temperature in the winter season which is preferred to this parasite as mentioned in Table (7).

Table 7: Number of generations of the mistletoe parasite in the six hosts/year

Hosts Treatments	Guava	Ardib	Nabg	Lemon	Grapefruit	Mango	Mean
Winter	2	2	2	2	2	2	2
Summer	1	1	1	1	1	1	1
Year	3	3	3	3	3	3	3

Conclusions and Recommendations:

Conclusions:

The study concluded that:

1. The mistletoe parasitism was parasitizing mainly on guava, ardib, nabg, lemon, grapefruit and acacia.
2. Among the six hosts, acacia, nabg and lemon are the highest susceptible to this parasite while guava and grapefruit are the lowest susceptible.
3. The number of generations which obtained in the winter season (two generations) is more than which obtained in the summer season (only one generation).

Recommendation:

The study recommends that: the parasite must be controlled by integrated weed management viz usage more than one method of weed control methods.

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